In general, the parties involved in the micro on-line payments are Customer, Service Provider and Payment processor. The Micro electronic payment transactions can be explained in the following way.

Step 1 Customer proves his authentication and the payment processor issues micro payments.

Step 2: Customer pays the micro payments to the online service provider and gets the requested goods or services from them.

Step 3: Service provider deposit micro payments received from the customer to the payment processor and gets the money.

16.2.2 Macro electronic payment systems

Macro electronic payment systems support payments of higher value. The security requirements are more rigorous in macro payment systems because of huge money transactions. Banks will impose a minimum transaction overhead on macro payment systems. These transactional over heads for the usage of computationally expensive cryptographic operations prevent these payment systems to be used for the payment of small amounts. Some of the popular macro on-line payment systems are mentioned below

- Card based payment systems
- Electronic account transfer
- Electronic cash payment systems
- Mobile payment systems and internet payment systems

16.3 Card Based Payments Systems

Payment cards are plastic cards that enable cashless payments. They are simple embossed plastic card that authenticates the card holder on behalf of card issuing company, which allows the user to make use of various financial services. More than 90% of online payments are card based payments, at the same time other e-payment methods are also gaining importance now-a-days.

Based on the transaction settlement method there are three widely used card based payment systems. They are

1. Credit card based payment systems (pay later)
2. Debit card based payment systems (pay now)
3. Stored value card based payment systems (pay before)

16.3.1 Credit Card

Credit card is an electronic payment system normally used for retail transactions. A credit card enables the bearer to buy goods or services from a vendor, based on the cardholder's promise to the card issuer to payback the value later with an agreed interest. Every credit card account has a purchase limit set by the issuing bank or the firm. A credit card is different from a debit card where the credit card issuer lends money to customer instead of deducting it from customer's bank account instantly.

The term credit card was first mentioned in 1887 in the sci-fi novel "Looking Backward" by Edward Bellamy. The modern credit cards concept was

XII CompApp_English version CHAPTER 16.indd 224 23-02-2020 14:52:14